

**Predicting Mortality in Critically Ill Patients:
A Logistic Regression Analysis of the SUPPORT2 Dataset**

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Author Note

This analysis was completed as a mini project milestone for AAI-500. The full project will expand upon the methods and findings described herein. Correspondence regarding this report should be directed to Andre Hoth.

Abstract

This report presents a preliminary logistic regression analysis of the SUPPORT2 dataset, a clinical dataset comprising 9,105 critically ill patients across eight disease groups. The primary objective was to predict all-cause mortality within six months of hospital admission using a parsimonious set of demographic, physiological, and disease-related features. Following targeted data cleaning and exploratory data analysis, a logistic regression model with balanced class weighting was trained on 80% of the sample and evaluated on the remaining 20%. The model achieved an overall accuracy of 68.3%, with disease group — particularly Lung Cancer and MOSF with Malignancy — and physiological severity scores emerging as the strongest predictors of mortality. Findings are discussed in the context of model limitations and recommendations for the full project analysis.

Keywords: logistic regression, mortality prediction, critical illness, SUPPORT2, class imbalance

Introduction

Predicting mortality in critically ill patients is a fundamental challenge in clinical medicine and a well-established domain for statistical modeling. Early and accurate identification of patients at high risk of death enables clinicians to allocate resources, initiate timely interventions, and engage patients and families in informed decision-making. The SUPPORT2 dataset — derived from the Study to Understand Prognoses, Preferences, Outcomes and Risks of Treatment (Knaus et al., 1995) — provides a rich source of clinical data for this purpose, capturing 9,105 hospitalizations across five major U.S. medical centers in the early 1990s.

The dataset contains 48 variables spanning patient demographics, primary diagnosis, physiological measurements, laboratory values, and follow-up outcomes. The primary outcome variable, death, captures all-cause mortality within six months of hospital admission, reflecting the study's focus on medium-term prognosis rather than in-hospital survival alone. A separate variable, hospdead, captures the narrower subset of patients who died during the index hospitalization. Physician-estimated survival probabilities at two and six months (surv2m, surv6m) were excluded as model predictors to avoid circularity, as these estimates were generated from the same clinical assessments captured by other features.

This report documents a preliminary mini-project analysis intended to establish familiarity with the dataset and define the analytical approach for the full project. The scope is intentionally limited: a single cleaning pass, two exploratory visualizations, and one classification model. The central research question is: can a small set of clinical and demographic features accurately predict six-month mortality in critically ill patients? Statistical

methods applied in this analysis draw on generalized linear models and binary classification techniques as described in Agresti and Kateri (2022).

Data Cleaning and Preparation

Initial Assessment

The raw dataset comprised 9,105 observations and 48 variables. An audit of missing values revealed pervasive missingness across 33 of 48 variables, ranging from 0.3% to 62.0%. The most affected variable was *adlp* (activities of daily living — patient reported) at 62.0% missing ($n = 5,641$), followed by urine output (53.4%), glucose (49.4%), and blood urea nitrogen (47.8%). Columns missing more than 40% of values were dropped entirely, as imputing more than half of a variable's values would introduce more synthetic data than observed data, undermining analytical integrity.

Feature Selection

Following the high-missingness column drop, a focused feature set of ten variables was selected based on clinical relevance, low missingness, and representativeness across variable types. Selected features included: age, sex, primary disease group (*dzgroup*), number of comorbidities (*num.co*), APACHE Physiology Score (*aps*), Support Physiology Score (*sps*), mean arterial blood pressure (*meanbp*), heart rate (*hrt*), respiratory rate (*resp*), and body temperature (*temp*). The *aps* and *sps* are validated composite severity-of-illness scores constructed from multiple physiological measurements, making them particularly informative single predictors.

Missing Data Handling and Encoding

After feature selection, rows with any remaining missing values were dropped. Only two observations (0.02%) were lost, yielding a working dataset of 9,103 patients. This row-wise

deletion approach is a simplification appropriate for the mini-project scope; the full project will evaluate median imputation and k-nearest neighbors imputation as alternatives. The binary variable sex was encoded as male = 1, female = 0. The eight-category disease group variable was one-hot encoded, dropping one reference category (ARF/MOSF with Sepsis) to avoid multicollinearity, yielding seven binary indicator columns. The final modeling dataset contained 9,103 rows and 16 features.

Descriptive statistics revealed several implausible extreme values in physiological variables: mean arterial blood pressure ranged from 0 to 195 mmHg, heart rate from 0 to 300 bpm, and respiratory rate from 0 to 90 breaths per minute. These likely reflect measurement artifacts or end-of-life recordings rather than true physiological values. Outlier treatment was deferred to the full project but is noted as a data quality concern.

Exploratory Data Analysis

Class Distribution

The target variable death exhibited substantial class imbalance: 6,201 patients (68.1%) died within the six-month follow-up window, while 2,904 (31.9%) survived. This imbalance has direct implications for model evaluation: a naive classifier predicting death for every patient would achieve 68.1% accuracy without learning any meaningful pattern. Consequently, accuracy alone is an insufficient evaluation metric for this dataset, necessitating examination of precision, recall, and F1-score by class.

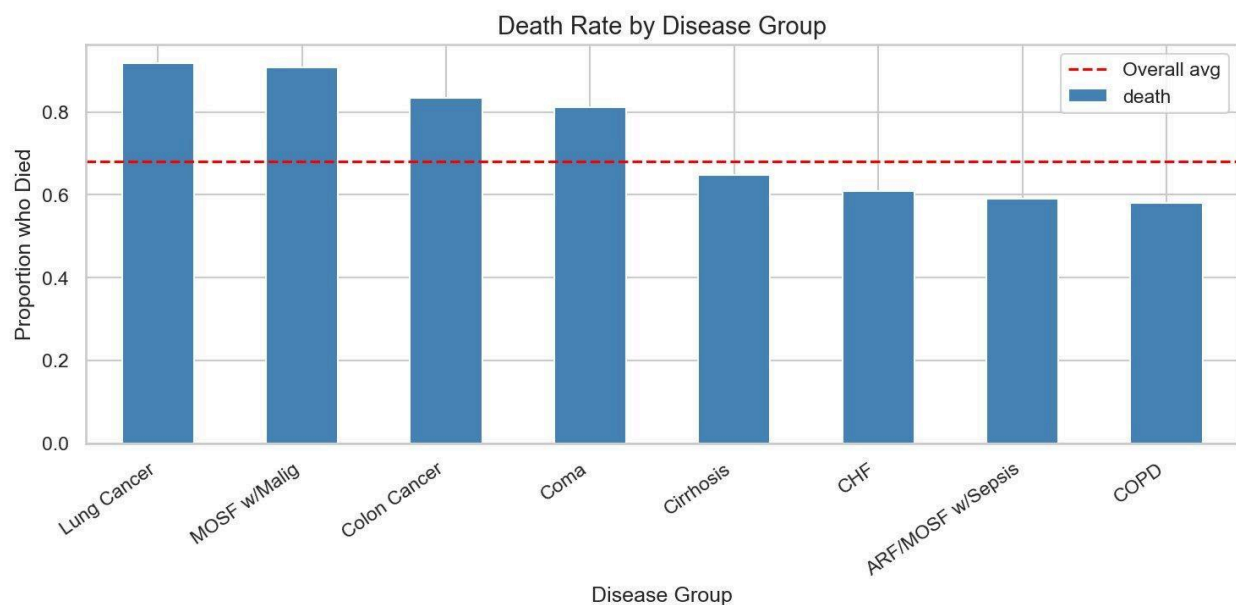
Death Rate by Disease Group

Mortality varied substantially across the eight primary disease groups, ranging from 58.0% among COPD patients to 91.7% among Lung Cancer patients — a spread of approximately 33.7 percentage points relative to the overall average of 68.1% (see Figure 1).

Lung Cancer and MOSF with Malignancy (90.9%) exhibited the highest mortality rates, both substantially above the overall average. In contrast, COPD, ARF/MOSF with Sepsis (59.1%), and congestive heart failure (CHF; 60.9%) fell below the average, representing comparatively more survivable conditions within this critically ill population. This variation suggests disease group is a meaningful predictor of mortality, warranting formal chi-square testing in the full project.

Figure 1

Six-month mortality rate by primary disease group



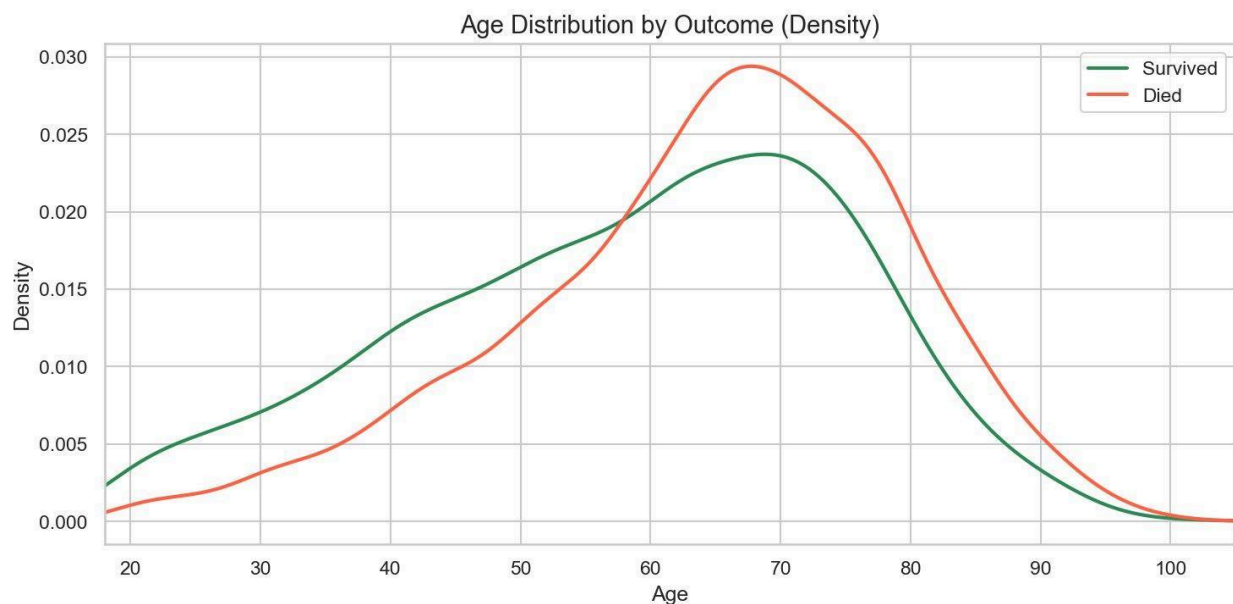
Note. The dashed red line indicates the overall sample mortality rate of 68.1%. Disease groups are ordered from highest to lowest mortality rate. ARF/MOSF = acute respiratory failure/multiple organ system failure; CHF = congestive heart failure; COPD = chronic obstructive pulmonary disease; MOSF w/Malig = multiple organ system failure with malignancy.

Age Distribution by Outcome

Kernel density estimation plots of age by outcome revealed similar distributions across both groups, with deceased patients skewing slightly older (peak density approximately age 70) compared to survivors (peak density approximately age 68; see Figure 2). The distributions crossed at approximately age 47, below which survivors were proportionally better represented. The modest separation between curves is consistent with the weak positive correlation observed between age and mortality ($r = 0.179$), indicating that age contributes predictive signal but is insufficient as a standalone predictor. The kernel density estimation approach was selected over overlapping histograms to avoid visual misinterpretation arising from class imbalance.

Figure 2

Kernel density estimates of age distribution by six-month mortality outcome



Note. Density curves are normalized independently for each outcome group, allowing direct comparison of distributional shape regardless of class size. The crossover point at approximately age 47 indicates the age below which survivors are proportionally better represented.

Feature Correlations

Pearson correlations between numeric features and the binary outcome variable ranged from $r = -0.031$ (meanbp) to $r = 0.179$ (age). Severity scores aps ($r = 0.157$) and sps ($r = 0.156$) demonstrated the strongest correlations among non-demographic features, consistent with their clinical design as mortality risk instruments. All individual correlations were weak, suggesting that predictive power resides in the combination of features rather than any single variable — a pattern well-suited to multivariate logistic regression.

Model Selection

Logistic regression was selected as the classification model for this analysis on the basis of three considerations. First, interpretability: logistic regression coefficients directly quantify the direction and magnitude of each feature's contribution to the log-odds of death, facilitating clinical interpretation. Second, appropriateness: with a binary outcome variable and a mix of continuous and categorical predictors, logistic regression represents the canonical modeling approach for binary response variables as described in Agresti and Kateri (2022, Chapter 7). Third, baseline utility: logistic regression provides a well-understood performance benchmark against which more complex models — such as decision trees or random forests — can be compared in the full project.

The 9,103-observation dataset was partitioned into training (80%, $n = 7,282$) and test (20%, $n = 1,821$) sets using stratified random sampling to preserve the 68.1% mortality rate in both partitions. All continuous features were standardized using z-score normalization fit exclusively on training data and applied to the test set, preventing data leakage. To address class imbalance, the model was trained with balanced class weighting (`class_weight = 'balanced'`), which adjusts the loss function to penalize misclassification of the minority class (survivors)

proportionally to its underrepresentation. The optimizer was permitted up to 1,000 iterations to ensure convergence.

Model Analysis

Overall Performance

The logistic regression model achieved an overall accuracy of 68.3% on the held-out test set ($n = 1,821$), marginally above the 68.1% naive baseline. While the accuracy figure alone suggests limited improvement over chance, disaggregation by class reveals a more nuanced picture.

Classification Performance by Class

For the deceased class (majority), the model achieved a precision of 0.86 and a recall of 0.64, yielding an F1-score of 0.73. When the model predicted death, it was correct 86% of the time; however, it failed to identify 36% of patients who actually died. For the survivor class (minority), precision was 0.50 and recall was 0.78, yielding an F1-score of 0.61. The model correctly identified 78% of survivors but generated substantial false positives — predicting death for patients who survived. Full classification metrics are reported in Table 1.

Table 1

Classification Report: Logistic Regression Model Performance on Test Set ($n = 1,821$)

	Precision	Recall	F1-score	Support
Survived	0.50	0.78	0.61	581
Died	0.86	0.64	0.73	1,240
Accuracy			0.68	1,821
Macro average	0.68	0.71	0.67	1,821

Weighted average	0.75	0.68	0.69	1,821
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Note. Precision = proportion of positive predictions that were correct. Recall = proportion of actual positives correctly identified. F1-score = harmonic mean of precision and recall. Support = number of observations in each class. Macro average = unweighted mean across classes.

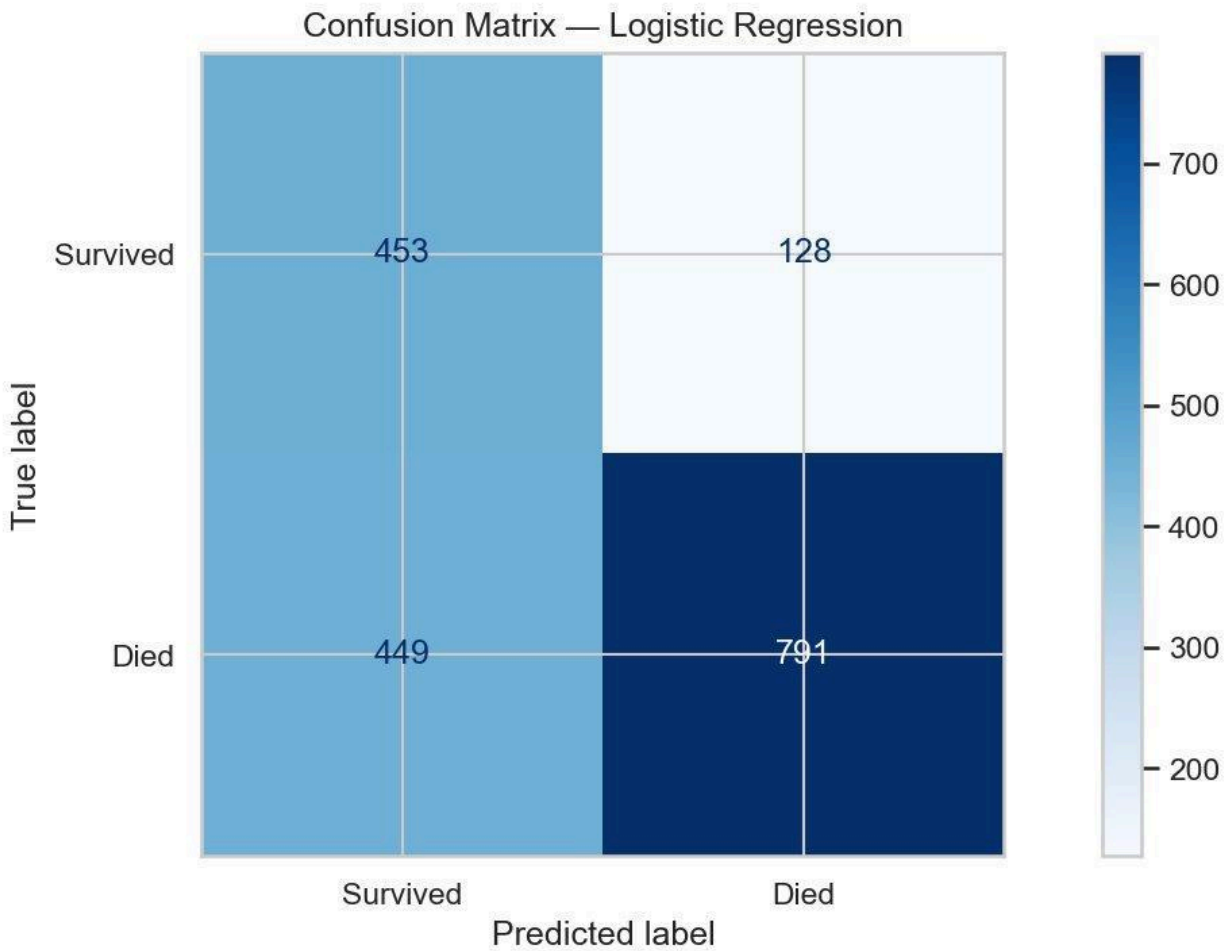
Weighted average = mean weighted by class support.

Confusion Matrix

Of 1,821 test observations, the model correctly classified 453 survivors (true negatives) and 791 deceased patients (true positives; see Figure 3). Error cells included 128 survivors predicted to die (false positives) and 449 deceased patients predicted to survive (false negatives). The 449 false negatives represent the most clinically significant error type — patients whose mortality risk was underestimated. The elevated false negative count is a direct consequence of balanced class weighting, which increased survivor recall at the cost of deceased recall. In clinical contexts where missed deaths carry higher consequences than unnecessary interventions, this tradeoff is defensible but warrants threshold optimization in the full project.

Figure 3

Confusion matrix for logistic regression predictions on the held-out test set



Note. Rows represent true outcome labels; columns represent model-predicted labels. Cell color intensity corresponds to the count of observations. Correct classifications appear on the main diagonal (top-left: true negatives = 453; bottom-right: true positives = 791). Off-diagonal cells represent misclassifications (top-right: false positives = 128; bottom-left: false negatives = 449).

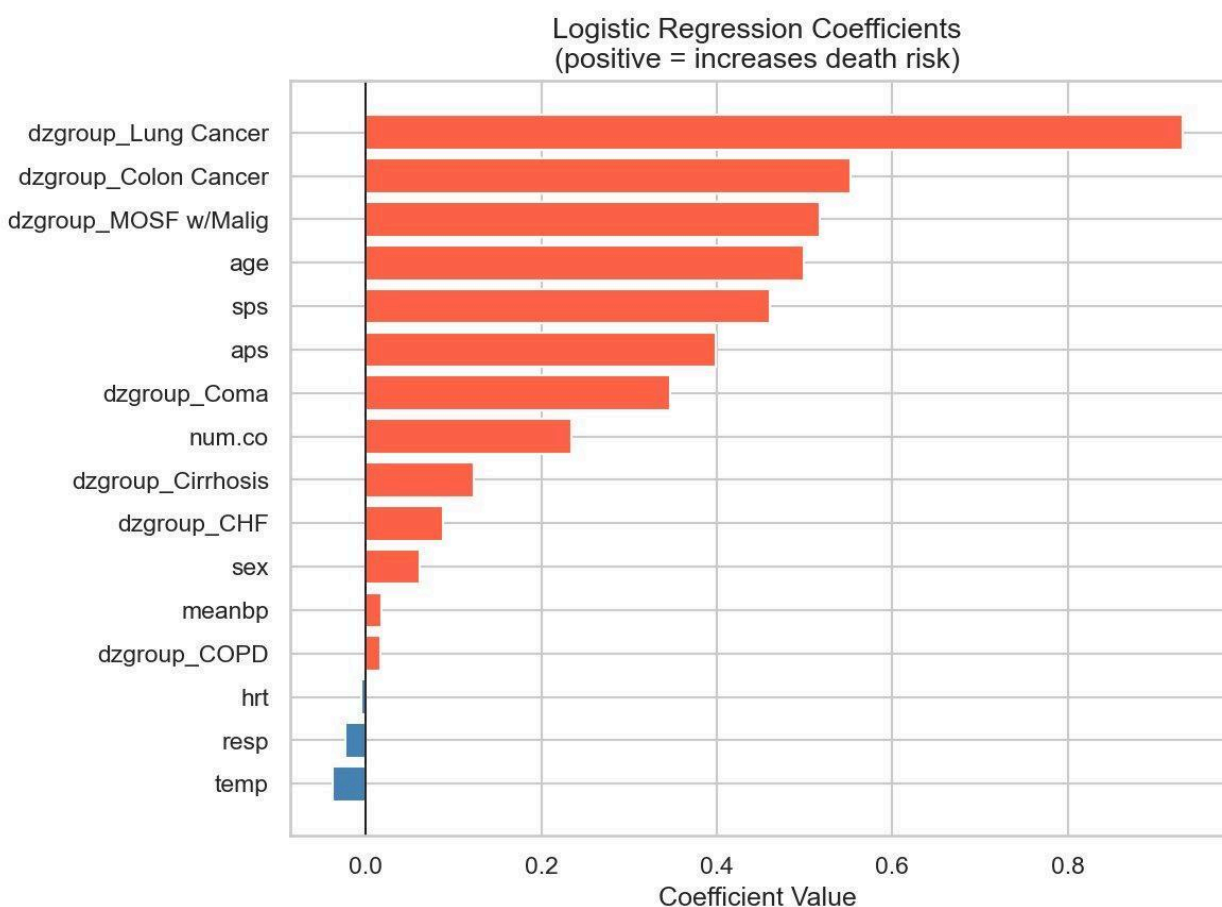
Feature Importance

Examination of standardized logistic regression coefficients identified disease group as the dominant predictor class (see Figure 4). Lung Cancer demonstrated the largest positive coefficient by a substantial margin, followed by Colon Cancer and MOSF with Malignancy — consistent with the mortality rates observed in the EDA. Among continuous features, age, sps,

and aps contributed meaningful positive coefficients, confirming the hypothesized predictive role of severity scores and patient age. Temperature and respiratory rate exhibited small negative coefficients, suggesting that lower values of these variables — potentially indicative of physiological deterioration — are associated with increased mortality risk, though the magnitude of these effects was negligible.

Figure 4

Standardized logistic regression coefficients by feature



Note. Coefficients are derived from a logistic regression model with balanced class weighting fit on the training set ($n = 7,282$). Positive coefficients (red) indicate features associated with

increased probability of death; negative coefficients (blue) indicate features associated with decreased probability of death. All features were standardized prior to model fitting. The reference category for disease group is ARF/MOSF with Sepsis. aps = APACHE Physiology Score; sps = Support Physiology Score; meanbp = mean arterial blood pressure; hrt = heart rate; resp = respiratory rate; temp = body temperature; num.co = number of comorbidities.

Conclusion and Recommendations

This preliminary analysis demonstrates that a parsimonious logistic regression model trained on ten clinical features can classify six-month mortality in critically ill patients at a rate modestly exceeding the naive baseline, with disease group — particularly Lung Cancer and MOSF with Malignancy — and physiological severity scores emerging as the most influential predictors. The analysis confirms the utility of the SUPPORT2 dataset for mortality prediction research and establishes a foundational understanding of its structure, missingness patterns, and key variable relationships.

Several limitations constrain the current findings. First, the cleaning strategy relied on row-wise deletion and excluded variables with high missingness, potentially discarding clinically informative features such as albumin (37% missing), a well-established mortality biomarker. Second, the single train/test split does not provide stable performance estimates; cross-validation would yield more reliable metrics. Third, implausible extreme values in physiological variables were retained without treatment, which may have introduced noise into the model. Fourth, classification threshold optimization was not performed; the default 0.5 threshold may not reflect the clinical cost asymmetry between false negatives and false positives.

For the full project, the following enhancements are recommended. Missing data should be addressed through median or k-nearest neighbors imputation, recovering informative

variables including albumin. Model comparisons should include decision tree and random forest classifiers to evaluate whether nonlinear methods improve predictive performance (Agresti & Kateri, 2022, Chapter 8). Classification threshold optimization via receiver operating characteristic curve analysis should be conducted on a dedicated validation set to minimize clinically costly false negatives. Outlier treatment and formal chi-square testing of disease group mortality differences should be incorporated into the cleaning and EDA phases respectively. Finally, k-fold cross-validation should replace the single train/test split to produce more stable and generalizable performance estimates.

References

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Appendix A

Jupyter Notebook Technical Details

The complete code and output for this analysis was produced in a Jupyter Notebook (support2_mini_project.ipynb) using Python 3.10. The notebook is organized into six sections corresponding to the structure of this report: Setup and Imports, Introduction and Data Loading, Data Cleaning and Preparation, Exploratory Data Analysis, Model Selection, and Model Analysis. The notebook will be submitted via the project GitHub repository as part of the full project deliverable.

Library versions used in this analysis: pandas 3.0.2, scikit-learn 1.8.0, matplotlib 3.10.9, seaborn 0.13.2. Random state was set to 42 throughout to ensure reproducibility of the train/test split and model fitting.